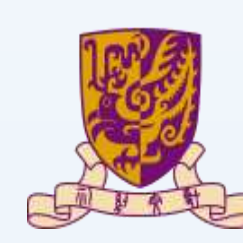




PRISM: Probing Reasoning, Instruction, and Source Memory in LLM Hallucinations

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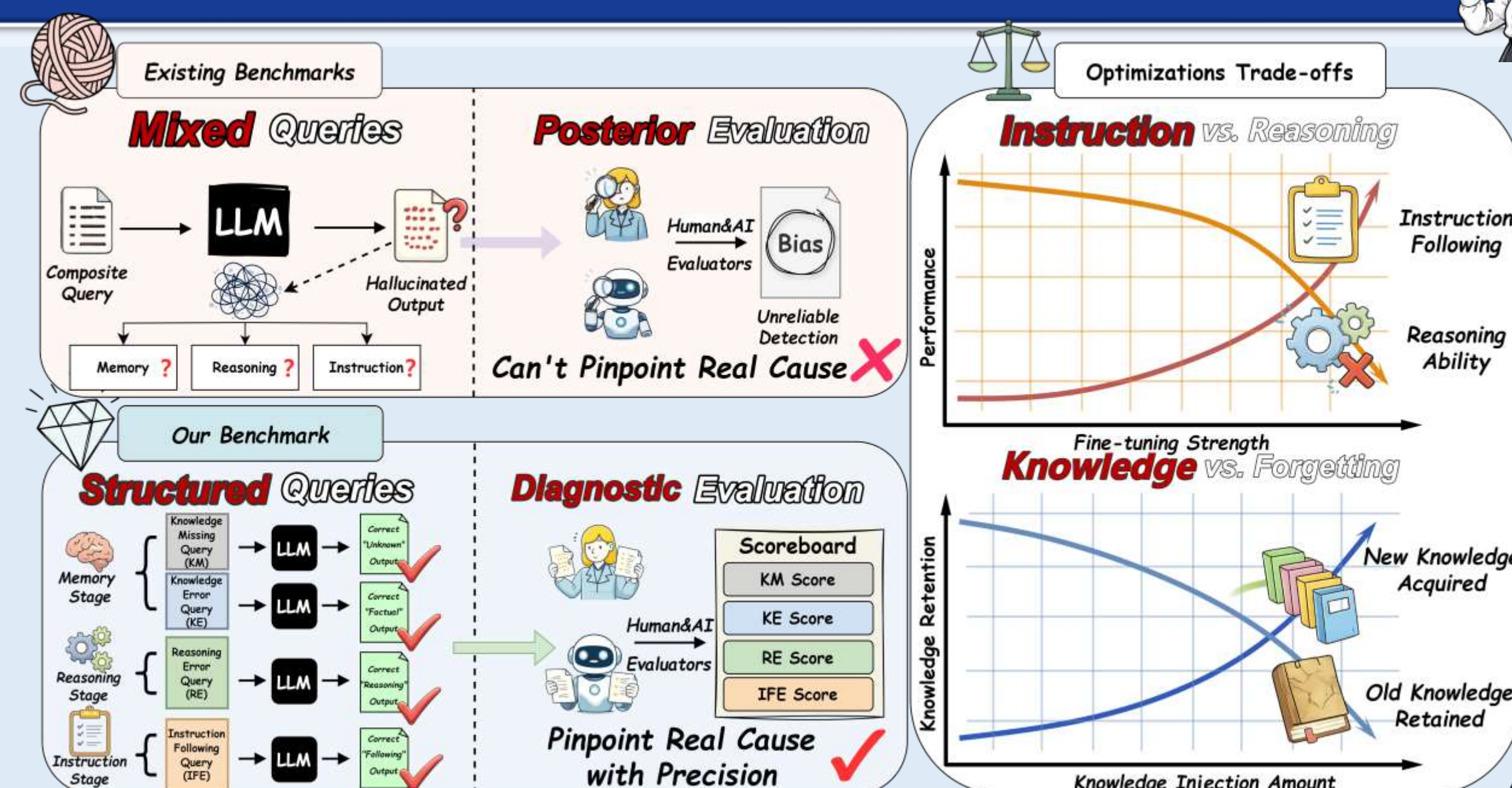
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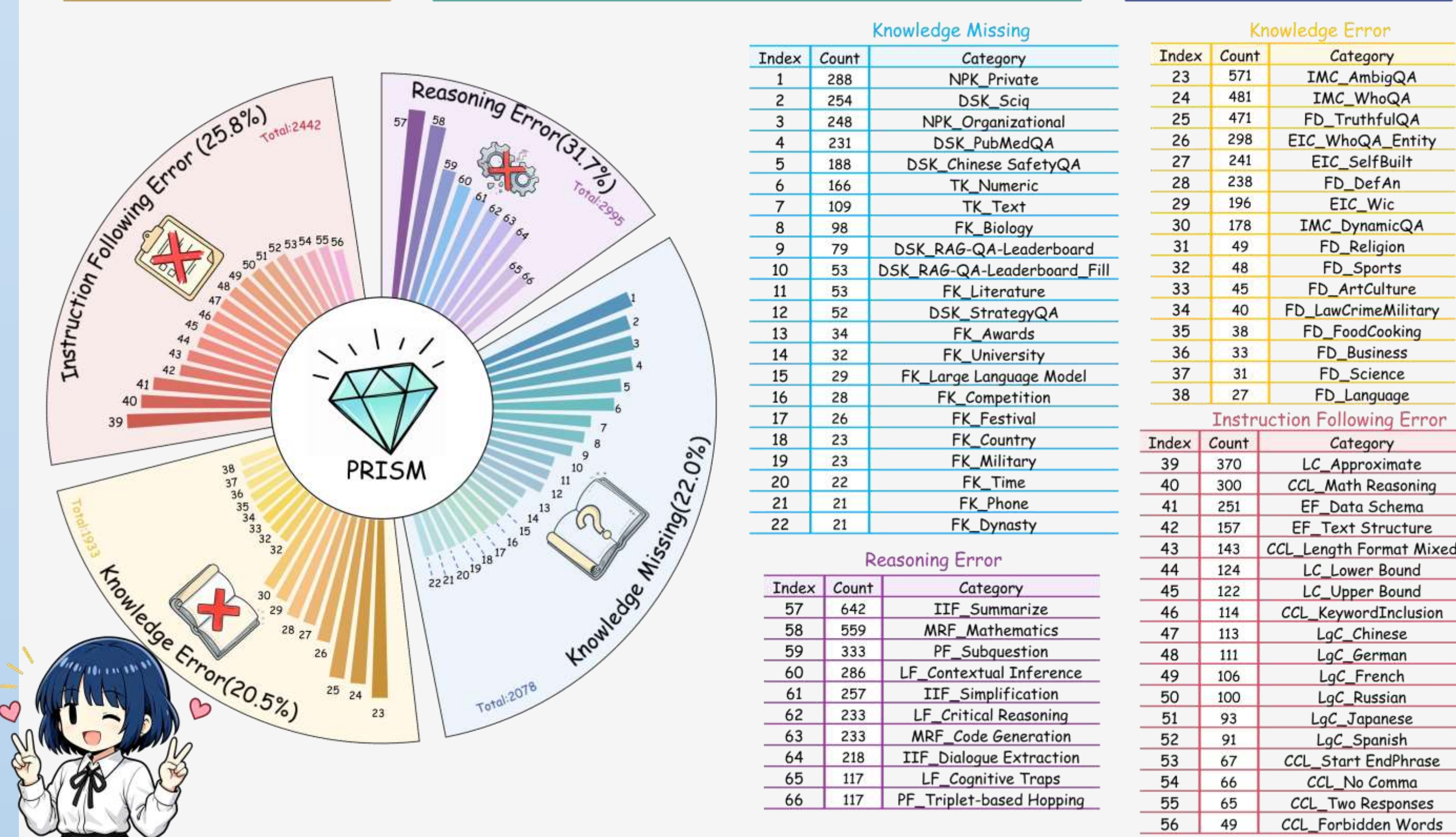
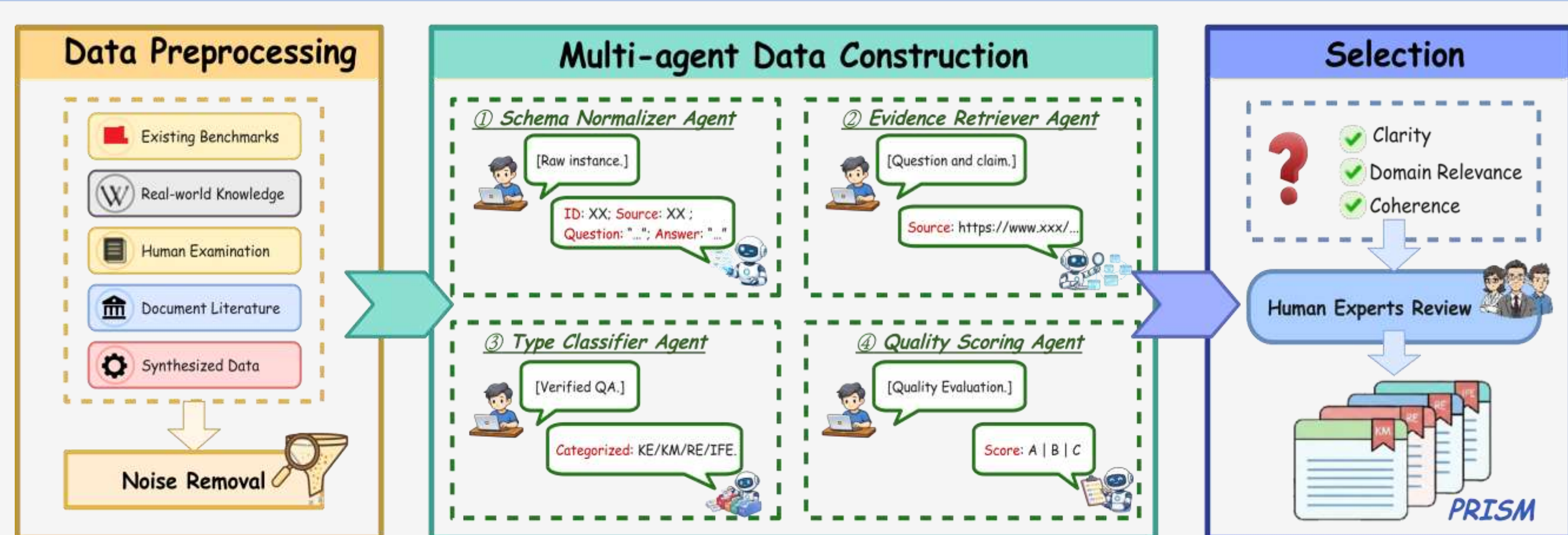
INTRODUCTION & MOTIVATION

- **Hallucination Attribution Gap:** Existing benchmarks quantify hallucination severity, but provide limited insight into why models fail during generation.
- **Challenge:** Mixed queries and final output scoring blur the boundary between memory retrieval, reasoning, and instruction following failures.
- **Our Approach:** PRISM introduces a stage aware diagnostic benchmark that probes incorrect knowledge, missing knowledge, reasoning errors, and instruction violations across 9,448 instances and 65 tasks.



PRISM

- **Construction Pipeline.** (i) Data Preprocessing: collect and clean raw sources. (ii) Multi-agent Data Construction: normalize schemas, retrieve evidence, classify types, and score quality. (iii) Human Selection: filter samples by clarity, relevance, and coherence.
- **Benchmark Dimensions.** Knowledge Error: incorrect stored knowledge; Knowledge Missing: absent required knowledge; Reasoning Error: failed logic over available facts; Instruction Following Error: violation of user constraints despite sufficient capability.



EXPERIMENT

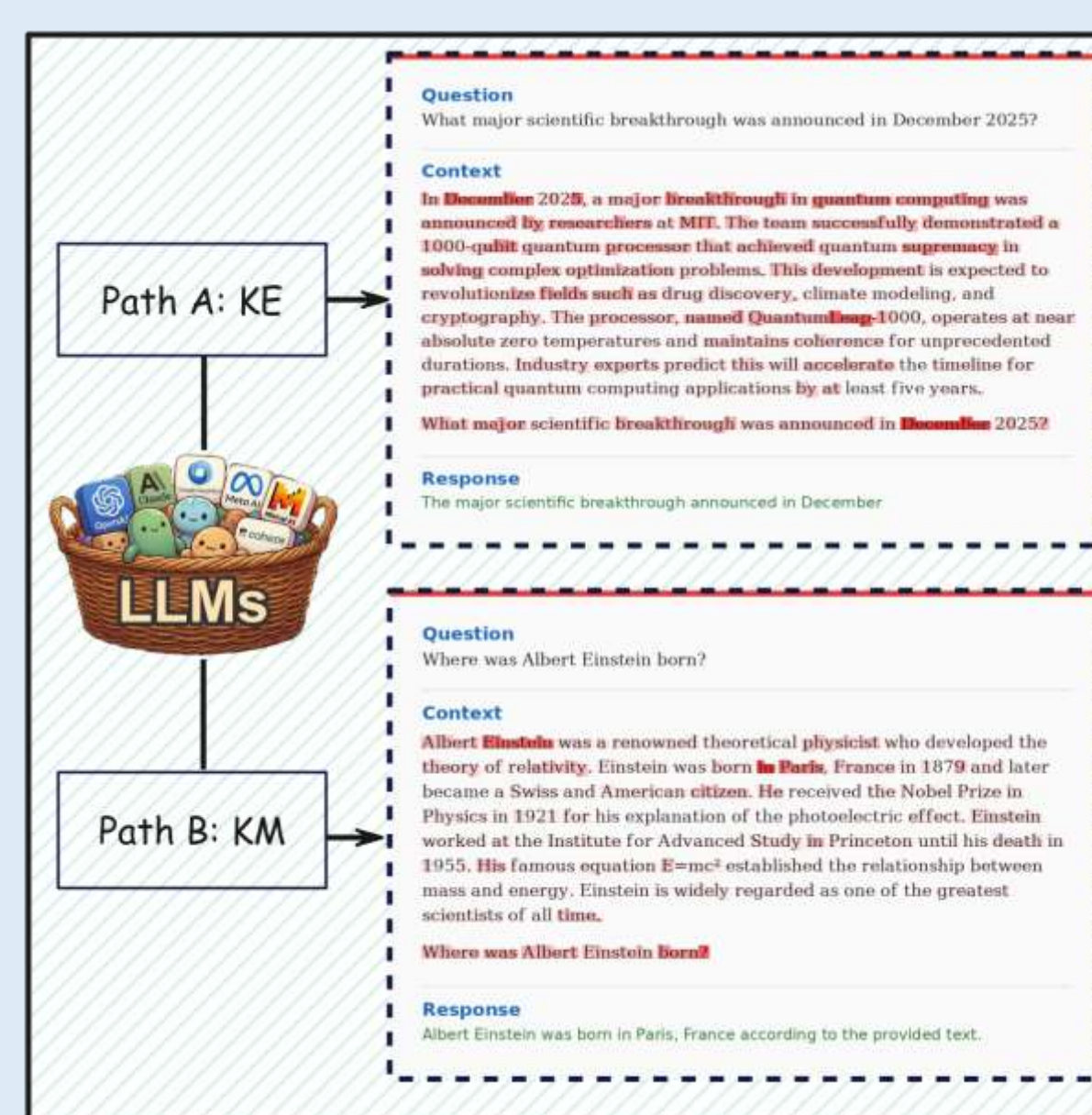
- **Main Experiment.** We evaluate representative proprietary and open-source LLMs on PRISM, reporting hallucination rates across four benchmark dimensions and the overall H-Score.

Type	Model	Size	Think	KE	KM	RE	IFE	H-Score
(Lower the Better)								
Proprietary LLMs	GPT-5.2	-	✓	17.83%	5.57%	24.67%	15.35%	15.85%
	GPT-5.1	-	✓	25.83%	11.46%	26.01%	25.62%	22.23%
	GPT-4o-20241120	-	✗	19.36%	8.90%	29.72%	16.78%	18.69%
	Gemini-3-Pro	-	✓	16.31%	12.80%	17.19%	10.85%	14.29%
	Gemini-3-Flash	-	✓	16.48%	8.12%	21.59%	16.00%	15.55%
	Gemini-2.5-Pro	-	✓	25.33%	8.99%	21.63%	13.84%	17.45%
	Gemini-2.5-Flash	-	✓	19.29%	8.74%	31.58%	15.51%	18.78%
	Claude-Opus-4.5	-	✓	13.80%	6.35%	19.68%	15.77%	13.90%
	Claude-Sonnet-4.5	-	✓	16.11%	7.03%	23.53%	16.87%	15.89%
	Claude-Haiku-4.5	-	✓	18.13%	7.24%	25.59%	18.15%	17.28%
Open-source LLMs	Grok-4.1	-	✗	19.20%	17.03%	34.94%	20.04%	22.80%
	Grok-4-0709	-	✗	18.35%	14.57%	30.97%	15.74%	19.91%
	DeepSeek-V3.2	685B	✗	17.99%	7.94%	28.31%	14.31%	17.14%
	DeepSeek-R1	671B	✓	18.87%	10.13%	28.04%	11.97%	17.25%
	DeepSeek-R1-Distill-32B	32B	✓	18.14%	11.27%	30.40%	18.71%	19.63%
	Qwen3-235B-Instruct	235B	✓	17.37%	7.00%	25.89%	17.87%	17.03%
	Qwen2.5-72B-Instruct	72B	✗	18.56%	8.61%	27.68%	18.43%	18.32%
	GLM-4.5	355B	✓	18.85%	10.70%	29.34%	15.06%	18.49%
	GLM-4	32B	✗	26.20%	16.17%	35.57%	25.34%	25.82%
	Llama-4-Scout	17B	✗	23.67%	7.58%	32.19%	14.17%	19.40%

- **Mitigation Trade-offs.** In-context learning brings only marginal changes, while instruction tuning improves KE and IFE but may weaken RE. In contrast, reasoning tuning reduces RE but worsens KE, KM, and IFE.

Model	Dataset	Base	ICL		Instruct. Reason.	
		1-shot	0-shot	3-shot	1-shot	1-shot
Llama3.1-8B	KE	43.15	43.88 (+0.73)	43.04 (-0.11)	38.61 (-4.54)	67.02 (+23.87)
	KM	11.95	13.28 (+1.33)	11.79 (-0.16)	11.78 (-0.17)	29.56 (+17.61)
	IFE	35.28	35.53 (+0.25)	35.03 (-0.25)	33.76 (-1.52)	61.87 (+26.59)
	RE_Math	94.99	95.71 (+0.72)	94.81 (-0.18)	95.17 (+0.18)	89.09 (-5.90)

Model	Task Type	ActN	AttnSp	GFLOPs
Qwen3-4B	IFE (Non-Refusal)	7.26 M	0.0191	215.7
	IFE (Refusal)	7.72 M	0.0175	229.3
LLama3.1-8B	IFE (Non-Refusal)	10.41 M	0.0148	468.9
	IFE (Refusal)	11.06 M	0.0137	498.4
Qwen3-1.7B	IFE (Non-Refusal)	4.73 M	0.0208	91.7
	IFE (Refusal)	5.03 M	0.0191	97.5
Qwen3-14B	IFE (Non-Refusal)	13.58 M	0.0169	755.0
	IFE (Refusal)	14.44 M	0.0154	802.6



Hallucination Mechanisms. KE is mainly driven by dominant parametric priors that suppress evidence use, while KM is more vulnerable to misleading context. Avoiding IFE through refusal also requires higher activation and computation, making safe responses more costly.

CONCLUSION & LIMITATIONS

- PRISM supports targeted hallucination diagnosis and reveals trade-offs behind mitigation strategies. Future work will extend to multimodal and broader multilingual settings, with rolling updates for evolving and long-tail knowledge.



See Where Hallucinations Begin.
Scan to explore PRISM and connect with us!

